STUDENT BOOK ANSWERS

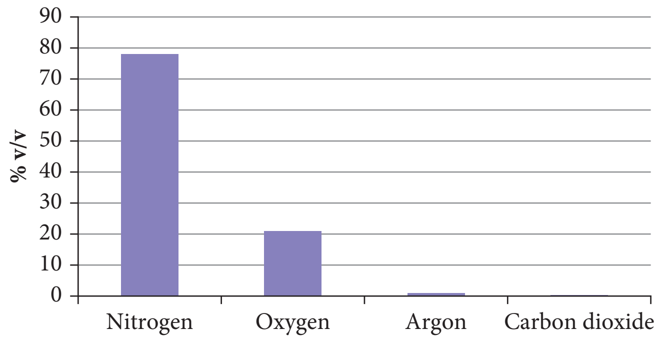
Chapter 8 Gases

Question set 8.1

1 Gas pressure is caused by the collisions of gas molecules with the walls of the container.

2 Pascal and atmosphere (Pa and atm)

3 a



b Argon is present at 0.93% v/v. In 100 L of air there would be 0.93 L of argon.

4 Height above sea level; temperature; weather conditions

5 a 1 atm = 101.3 KPa so $\frac{75.5 \text{ kPa}}{101.3} = 0.745$ atm

b $\frac{6.83}{101325} = 0.0000674$ atm

c $\frac{652}{760} = 0.858$ atm

6 a $1.03 \times 101.325 = 104.4$ kPa

b $0.45 \times 101.325 = 45.6$ kPa

c $\frac{820}{760} \times 101.325 = 109$ kPa

7 a No, the vacuum cleaner relies on the pressure drop between the outside and the inside of the cleaner.

b Yes, the siphon relies on liquid cohesion and gravity to work.

c No, there is no oxygen to burn.

d Yes, electromagnetic forces do not rely on an atmosphere as evidenced by the detection of the magnetic field in space around Earth.

Question set 8.2

1 • Gases consist of molecules (except the noble gases which consist of atoms) that move in continual random straight-line motion.

• The average distance between gas molecules is very large compared to the size of the molecule.

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- Intermolecular forces between molecules is negligible.
- All collisions of gas molecules are perfectly elastic, which means there is no net energy loss during these collisions.
- Pressure is due to collisions of the molecules with the walls of the container.
- Temperature is a measure of the average kinetic energy of the molecules.

2 No net loss of energy during collisions.

3 The theoretical lowest possible temperature.

4 **a** Gases can be compressed as there are very large distances between molecules.

b Gases can be mixed rapidly as there are very large distances between the molecules and the gas particles have rapid, random motion.

c Gases have low density as there are very large distances between molecules

5 An increase in temperature increases the average kinetic energy of the gas particles. The particles will hit the sides of the balloon with more force and more frequently. This increases the pressure inside the balloon. Eventually the pressure increases the size of the balloon so much that it bursts.

6 If the collisions were not elastic energy would be lost when the gas particles collided. The gas particles would slow down and the pressure in the balloon would decrease. This would cause the volume to also decrease.

7 The kinetic energy is the same for both gases. KE = $\frac{1}{2}mv^2$. However, hydrogen has less mass than oxygen, and therefore travels faster.

8 Ideal gas is a gas that obeys the relationship linking volume, mass pressure and temperature perfectly. The ideal gas obeys the ideal gas law $PV = nRT$. The real gas is the actual gas being examined.

9 **a** $237 + 273 = 510\,K$

b $-25 + 273 = 248\,K$

c $120 - 273 = -153°C$

d $345 - 273 = 72°C$

10 Gas will occupy a larger volume at higher temperatures. The petrol in the can on the hot day filled the air gap with petrol fumes. When sealed the pressure inside the can equalled the pressure outside. When the can cooled overnight some of the petrol fumes condensed back to liquid petrol. As the can was sealed no air could enter the can. The pressure inside the can decreased and was now less than the pressure outside the can. The can was then partially crushed.

11 Real gases obey the ideal gas law when there are no interactions between the particles. At high temperature the particles have higher kinetic energy and move more quickly. There is no time for the intermolecular forces to occur. At low pressure the particles are further apart and are less likely to have intermolecular interactions between particles.

(At low temperatures the particles slow down and may allow the intermolecular forces to have a more significant effect. This will also happen with high pressure as the particles are forced together and intermolecular forces can come into play. This also occurs when the gases are polar as hydrogen bonding is the strongest intermolecular force.)

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12 A. The gas molecules have the same average kinetic energy. However, heavier molecules will move slower than the lighter molecules, as shown by the relationship: $KE = \frac{1}{2}mv^2$

$M(N_2) = 28$, $M(F_2) = 38$, $M(CO_2) = 44$

Nitrogen will move the fastest.

13 Gases will diffuse to fill the container so the bottle above the liquid will always be filled with gas.

14 The one that would travel the distance the fastest would be the lighter weight one. The two gases would have the same kinetic energy. $KE = \frac{1}{2}mv^2$. So heavier gases would move slower, and lighter gas molecules would move faster. Ammonia is lighter than chlorine so would be smelt first.

Worked example 8.1

a $P_1 = 94.2\,\text{kPa}$ $\quad P_2 = 101.3\,\text{kPa}$

$V_1 = 4.42\,\text{L}$ $\quad V_2 = ?$

$P_1V_1 = P_2V_2$

$94.2 \times 4.42 = 101.3 \times V_2$

$$V_2 = \frac{P_1 \times V_1}{P_2}$$

$$V_2 = \frac{94.2 \times 4.42}{101.3}$$

$$= 4.11\,\text{L}$$

b $P_1 = 1\,\text{atm}$ $\quad P_2 = ?\,\text{atm}$

$V_1 = 210\,\text{mL}$ $\quad V_2 = 150\,\text{mL}$

$P_1V_1 = P_2V_2$

$1 \times 210 = P_2 \times 0.150$

$$P_2 = \frac{P_1 \times V_1}{V_2}$$

$$P_2 = \frac{1 \times 210}{150}$$

$$= 1.4\,\text{atm}$$

c $P_1 = 522\,\text{kPa}$ $\quad P_2 = 101.3\,\text{kPa}$

$V_1 = 55\,\text{mL}$ $\quad V_2 = ?$

$P_1V_1 = P_2V_2$

$522 \times 55 = 101.3 \times V_2$

$$V_2 = \frac{P_1 \times V_1}{P_2}$$

$$V_2 = \frac{522 \times 55}{101.3}$$

$$= 283\,\text{mL}$$

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d $P_1 = 73.3\,\text{kPa}$ $\qquad P_2 = ?\,\text{kPa}$

$V_1 = 2.5\,\text{L}$ $\qquad V_2 = 7.2\,\text{L}$

$P_1V_1 = P_2V_2$

$$P_2 = \frac{P_1 \times V_1}{V_2}$$

$73.3 \times 2.5 = P_2 \times 7.2$

$$P_2 = \frac{73.3 \times 2.5}{7.2}$$

$= 2.5\,\text{kPa}$

Experiment 8.1: Pressure and volume

Results

When the bottle is squeezed, the 'diver' sinks. When the bottle is released, the 'diver' floats.

Discussion

Boyle's law states that pressure is inversely proportional to volume. The increase in pressure when the bottle is squeezed compresses the air. The volume occupied by the gas decreases. This increases the density of the 'diver'. The 'diver' sinks. When the bottle is released, the pressure inside the bottle decreases so the volume of air expands. This decreases the density of the 'diver' and the 'diver' rises.

Worked example 8.2

a $$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$V_1 = 3.50\,\text{mL}$ $\qquad V_2 = ?$

$T_1 = 20°\text{C}$ $\qquad T_2 = 80°\text{C}$

$= 293\,\text{K}$ $\qquad = 353\,\text{K}$

$$V_2 = \frac{V_1 \times T_2}{T_1}$$

$$V_2 = \frac{3.50 \times 353}{293} = 4.2\,\text{mL}$$

b i $$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$V_1 = 1.7\,\text{L}$ $\qquad V_2 = ?$

$T_1 = 25°\text{C}$ $\qquad T_2 = 100°\text{C}$

$= 298\,\text{K}$ $\qquad = 373\,\text{K}$

$$V_2 = \frac{V_1 \times T_2}{T_1}$$

$$V_2 = \frac{1.7 \times 373}{298} = 2.1\,\text{L}$$

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ii $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$V_1 = 1.7\,\text{L}$ $\quad V_2 = 0.80$

$V_1 = 25°\text{C}$ $\quad T_2 = ?$

$= 298\,\text{K}$

$$T_2 = \frac{V_2 \times T_1}{V_1}$$

$$T_2 = \frac{0.80 \times 298}{1.7} = 140\,\text{K} = -132°\text{C}$$

c $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$V_1 = 60.0\,\text{mL}$ $\quad V_2 = ?$

$T_1 = 36°\text{C}$ $\quad T_2 = 0°\text{C}$

$= 309\,\text{K}$ $\quad = 273\,\text{K}$

$$V_2 = \frac{V_1 \times T_2}{T_1}$$

$$V_2 = \frac{60.0 \times 273}{309} = 53\,\text{mL}$$

Worked example 8.3

a $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$P_1 = 135\,\text{kPa}$ $\quad P_2 = 101.3\,\text{kPa}$

$V_1 = 202\,\text{mL}$ $\quad V_2 = ?$

$T_1 = 80°\text{C} = 353\,\text{K}$ $\quad T_2 = 0°\text{C} = 273\,\text{K}$

$$V_2 = \frac{P_1 \times V_1 \times T_2}{T_1 \times P_2}$$

$$V_2 = \frac{135 \times 202 \times 273}{353 \times 101.3}$$

$V_2 = 208\,\text{mL}$

b i $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$P_1 = 101.3\,\text{kPa}$ $\quad P_2 = 30\,\text{kPa}$

$V_1 = 24.5\,\text{L}$ $\quad V_2 = ?$

$T_1 = 25°\text{C} = 298\,\text{K}$ $\quad T_2 = -30°\text{C} = 243\,\text{K}$

$$V_2 = \frac{P_1 \times V_1 \times T_2}{T_1 \times P_2}$$

$$V_2 = \frac{101.3 \times 24.5 \times 243}{298 \times 30}$$

$V_2 = 67.5\,\text{L}$

ii $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$P_1 = 101.3\,\text{kPa}$ $P_2 = 0.80\,\text{kPa}$

$V_1 = 24.5\,\text{L}$ $V_2 = ?$

$T_1 = 25°\text{C} = 298\,\text{K}$ $T_2 = 27°\text{C} = 300\,\text{K}$

$$V_2 = \frac{P_1 \times V_1 \times T_2}{T_1 \times P_2}$$

$$V_2 = \frac{101.3 \times 24.5 \times 300}{298 \times 0.80}$$

$V_2 = 3123\,\text{L}$

c $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$P_1 = 200\,\text{kPa}$ $P_2 = ?$

$V_1 = 20.0\,\text{L}$ $V_2 = 24.0\,\text{L}$

$T_1 = 15°\text{C} = 288\,\text{K}$ $T_2 = 40°\text{C} = 313\,\text{K}$

$$P_2 = \frac{P_1 \times V_1 \times T_2}{T_1 \times V_2}$$

$$P_2 = \frac{200 \times 20.0 \times 313}{288 \times 24}$$

$P_2 = 181\,\text{kPa}$

Question set 8.3

1 **a** Standard temperature and pressure: 0°C and 1.0 atm

b 25°C and 1.0 atm

2 **a** At a constant temperature the volume of a given mass of gas is inversely proportional to the applied pressure.

b At a constant pressure, the volume of a given mass of gas is proportional to the absolute temperature.

3 In winter the air temperature is colder. A given mass of gas will take up less volume. If air is not added to the tyres the tyre will be deflated, the edges of the tyre tread will wear out and the steering will feel heavy. The tyre may not grip the road as well resulting in more dangerous driving. In summer the air temperature is higher. The same amount of gas would increase the pressure in the tyre, so the weight of the car will concentrate on the centre of the tread, wearing that out, provide less area in contact with the road affecting safety and making steering too light. Some air has to be released to suit the conditions.

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4 If all conditions remain the same; a doubling of the temperature will result in a doubling of the volume. A 4 L balloon two thirds full has a volume of 2.7 L. If this was to double the volume would be 5.3 L. The balloon would burst.

5 a $P_1 = 1.05\,\text{atm}$ $\quad P_2 = 3\,\text{atm}$

$V_1 = 6.2\,\text{L}$ $\quad V_2 = ?$

$P_1V_1 = \text{P}_2\text{V}_2$

$$V_2 = \frac{P_1 \times V_1}{P_2}$$

$$V_2 = \frac{1.05 \times 6.2}{3}$$

$= 2.2\,\text{L}$

b $P_1 = 2.3\,\text{atm}$ $\quad P_2 = 1\,\text{atm}$

$V_1 = 24\,\text{L}$ $\quad V_2 = ?$

$P_1V_1 = P_2V_2$

$$V_2 = \frac{P_1 \times V_1}{P_2}$$

$$V_2 = \frac{2.3 \times 24}{1}$$

$= 55\,\text{L}$

6 a $$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$V_1 = 450\,\text{mL}$ $\quad V_2 = 200\,\text{mL}$

$T_1 = 100°\text{C}$ $\quad T_2 = ?$

$= 373\,\text{K}$

$$T_2 = \frac{V_2 \times T_1}{V_1}$$

$$T_2 = \frac{200 \times 373}{450} = 166\,\text{K} = -107°\text{C}$$

b $$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$V_1 = ?$ $\quad V_2 = 100\,\text{mL}$

$T_1 = 25°\text{C}$ $\quad T_2 = 50°\text{C}$

$= 298\,\text{K}$ $\quad = 323\,\text{K}$

$$V_1 = \frac{V_2 \times T_1}{T_2}$$

$$V_1 = \frac{100 \times 298}{323} = 92\,\text{mL}$$

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7 $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$P_1 = 1.2\,\text{atm}$ $P_2 = 1\,\text{atm}$

$V_1 = 10.0\,\text{L}$ $V_2 = ?$

$T_1 = 25°\text{C} = 298\,\text{K}$ $T_2 = 0°\text{C} = 273\,\text{K}$

$$V_2 = \frac{P_1 \times V_1 \times T_2}{T_1 \times P_2}$$

$$V_2 = \frac{1.2 \times 10.0 \times 273}{298 \times 1.0}$$

$V_2 = 11\,\text{L}$

8 $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$P_1 = 103\,\text{kPa}$ $P_2 = ?$

$V_1 = \text{X L}$ $V_2 = 1.1\,\text{X L}$

$T_1 = 15°\text{C} = 288\,\text{K}$ $T_2 = 200°\text{C} = 473\,\text{K}$

$$P_2 = \frac{P_1 \times V_1 \times T_2}{T_1 \times P_2}$$

$$P_2 = \frac{103 \times X \times 473}{288 \times 1.1X}$$

$P_2 = 154\,\text{kPa}$

9 $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

$P_1 = ?$ $P_2 = 100\,\text{atm}$

$T_1 = 25°\text{C} = 298\,\text{K}$ $T_2 = 300°\text{C} = 573\,\text{K}$

$$P_1 = \frac{P_2 \times T_1}{T_2}$$

$$P_2 = \frac{100 \times 298}{573}$$

$P_2 = 52\,\text{atm}$

10 a $P_1 = 101.3\,\text{kPa}$ $P_2 = 79.4\,\text{kPa}$

$V_1 = 430\,\text{m}^3$ $V_2 = ?$

$$V_2 = \frac{P_1 \times V_1}{P_2}$$

$$V_2 = \frac{101.3 \times 430}{79.4}$$

$= 548\,\text{m}^3$

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b With the decrease in pressure as the hurricane approaches the volume of gas inside a room will expand. If the windows are not left slightly ajar the room could burst.

Worked example 8.4

a $V = n \times 22.4 = 0.095 \times 22.4 = 2.13\,\text{L}$

b **i** $n = \dfrac{V}{22.4} = \dfrac{0.450}{22.4} = 0.02\,\text{mol}$

ii $m = nM = 0.02 \times 32 = 0.64\,\text{g}$

c **i** $n = \dfrac{m}{M} = \dfrac{10}{2} = 5\,\text{mol}$

$V = 5 \times 22.4 = 112\,\text{L}$

ii $n = \dfrac{m}{M} = \dfrac{10}{44} = 0.23\,\text{mol}$

$V = 0.23 \times 22.4 = 5.1\,\text{L}$

iii The volume is proportional to the number of moles, not the mass. 10 g of hydrogen is 5 moles and 10 g of carbon dioxide is 0.23 mol. The volume occupied by the carbon dioxide should be much less.

Worked example 8.5

a $PV = nRT$

$P = 75.0\,\text{kPa}$

$V = ?$

$n = 0.34$

$T = 100°\text{C} = 373\,\text{K}$

$R = 8.314\,\text{J}\,\text{K}^{-1}\,\text{mol}^{-1}$

$$V = \frac{n \times 8.314 \times T}{P}$$

$$V = \frac{0.34 \times 8.314 \times 373}{75}$$

$= 14\,\text{L}$

b $PV = nRT$

$m = ?$

$P = 1.05\,\text{atm}$

$V = 527\,\text{mL}$

$n = ?$

$T = 20°\text{C} = 293\,\text{K}$

$R = 0.082\,\text{L}\,\text{atm}$

$1.05 \times 0.527 = n \times 0.082 \times 293$

$$n = \frac{1.05 \times 0.527}{0.082 \times 293}$$
$$= 0.023\,\text{mol}$$

$$n = \frac{m}{M}$$

$M(O_2) = 32\,\text{g}$

$$0.023 = \frac{m}{32}$$

$m = 0.023 \times 32 = 0.736\,\text{g}$

c $PV = nRT$

$m = 4.00\,\text{g}$

$P = ?\,\text{kPa}$

$V = 7.50\,\text{L}$

$n = ?$

$T = 30°\text{C} = 303\,\text{K}$

$R = 8.314\,\text{J K}^{-1}\,\text{mol}^{-1}$

$$n = \frac{m}{M} = \frac{4}{4} = 1\,\text{mol}$$

$$P = \frac{n \times 8.314 \times T}{V} = \frac{1 \times 8.314 \times 303}{7.50} = 336\,\text{kPa}$$

Experiment 8.2: Measuring the molar mass of butane

Analysing the results

1 Assume: 3.3 grams butane

2 $n = \frac{m}{M} = 0.057\,\text{mol}$

3 This will depend upon the experiment temperature, 25°C taken for calculations

4 Assuming 1 atm, the vapour pressure $= 1 - 0.0312 = 0.9688\,\text{atm}$

5 Results will vary depending on the atmospheric pressure and the mass used

$PV = nRT$, assume volume in cylinder $= 1.4\,\text{L}$

$$n = \frac{PV}{RT} = \frac{0.9688 \times 1.4}{0.082 \times (273 + 25)} = 0.056\,\text{mol}$$

This was for 3.3 grams, $n = \frac{m}{M}$, $M = \frac{m}{n} = \frac{3.3}{0.056} = 58.9$

Discussion

1 This will depend upon students' results.

2 This will depend upon students' results.

3 Some of the pressure recorded was due to the water in the atmosphere not due to the presence of butane.

4 Students' answers will vary.

Question set 8.4

1 Avogadro's hypothesis is that equal volume of any gas, measured at the same temperature and pressure contain the same number of particles.

2 a The volume occupied by one mole (6.02×10^{23} particles) of a gas.

b $PV = nRT$

3 a R has different values depending on the units used. As the pressure and volume can be measured in different units such as atmospheres or mL rather than the SI units of kPa and L, a different R is required to reflect the different units.

b At low temperatures the particles slow down and may allow the intermolecular forces to affect their behaviour. This will also happen with high pressure as the particles are forced together and intermolecular forces can come into play. This also occurs when the gases are polar as the intermolecular force is stronger.

4 $PV = nRT$

$$V = \frac{nRT}{P} = \frac{1 \times 0.082 \times 773}{0.80} = 79.2\,\text{L}$$

5 $$n = \frac{PV}{RT} = \frac{20 \times 5.0}{8.314 \times 300} = 0.040\,\text{mol}$$

6 $$n = \frac{m}{M} = \frac{2.59}{28} = 0.0925\,\text{mol}$$

$$P = \frac{nRT}{V} = \frac{0.0925 \times 8.314 \times 303}{8} = 29\,\text{KPa}$$

7 $R = 0.082\,\text{L atm}$

$$V = \frac{nRT}{P} = \frac{0.200 \times 0.082 \times 293}{0.97} = 5.0\,\text{L}$$

8 $$n_1 = \frac{PV}{RT} = \frac{1.00 \times 1.00}{0.082 \times 273} = 0.04467\,\text{mol}.\ M = \frac{m}{n} = \frac{1.89}{0.04467} = 42.3$$

$$n_2 = \frac{1.25 \times 1.00}{0.082 \times 473} = 0.03223\,\text{mol}.\ M = nM = 0.03223 \times 42.3 = 1.36\,\text{g}$$

9 $$n = 0.00518\,\text{mol},\ M = \frac{m}{n} = \frac{0.20}{0.00518} = 38.6\ \text{Ar}$$

10 $PV = nRT$

$R = 0.082\,\text{L atm}$

$V = 5.0\,\text{L}$

$T = 27°\text{C} = 300\,\text{K}$

$P = 1\,\text{atm}$

Avogadro's law states that at equal volume of any gas, measured at the same temperature and pressure contain the same number of particles.

So n is the same for all. What is different is the mass.

$n = PV/RT = (1 \times 5)/(0.082 \times 300) = 0.20\,\text{mol}$

$M(H_2) = 2\,\text{g mol}^{-1}$, $m(H_2) = nM = 0.20 \times 2 = 0.4\,\text{g}$

$M(CO_2) = 44\,\text{g mol}^{-1}$ $m(CO_2) = nM = 0.20 \times 44 = 8.8\,\text{g}$

$M(\text{air}) = 29.0\,\text{g mol}^{-1}$ $m(\text{air}) = nM = 0.20 \times 29 = 5.8\,\text{g}$

The hydrogen balloon weighing 0.4 g is much lighter than air (at 5.8 g), so would rise. The carbon dioxide balloon weighs 8.8 g, which is heavier than air, so it would sink.

Worked example 8.6

a $4Mg(s) + 2NO_2(g) \rightarrow 4MgO(s) + N_2(g)$

$n = \dfrac{m}{M}$

$n(Mg) = \dfrac{m}{M} = \dfrac{4.86}{24.3} = 0.2\text{ mol}, n(N_2) = \dfrac{1}{4}\, n(Mg) = 0.05\,\text{mol}$

$n = \dfrac{V}{22.4}$

$V = 0.05 \times 22.4 = 1.1\,\text{L}$

b $C(s) + O_2(g) \rightarrow CO_2(g)$

$n = \dfrac{m}{M}$

$n(C) = \dfrac{10}{12} = 0.83\text{mol} \rightarrow n(O_2) = 0.83$

$n = \dfrac{V}{24.5}$

$V(CO_2)$ produced $= V(O_2)$ consumed $= 0.83 \times 24.5 = 20.4\,\text{L}$

c $PV = nRT$

$P = 1.3\,\text{atm}$

$V = 120\,\text{mL} = 0.120\,\text{L}$

$n = ?$

$R = 0.082\,\text{L atm}$

$T = 305\,\text{K}$

$PV = nRT, n(O_2) = \dfrac{PV}{RT} = \dfrac{1.3 \times 0.120}{0.082 \times 305} = 0.0062\,\text{mol}$

$n(KClO_3) = \dfrac{2}{3} \times n(O_2) = \dfrac{m}{M}$

$m = nM = 0.0062 \times \dfrac{2}{3} \times 122.6 = 0.51\,\text{g}$

0.51 g of potassium chlorate must be heated to give 120 mL of oxygen.

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d $2H_2O_2(l) \rightarrow O_2(g) + 2H_2O(l)$

$n(O_2) = \frac{V}{24.5} = 100/24.5 = 4.08$ mol. $n(H_2O_2) = 2\,n(O_2)$

$n = \frac{m}{M}$, $m = nM = 2 \times 4.08 \times 34 = 278$ g of hydrogen peroxide needed.

e $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

$PV = nRT$

$P = 110\,\text{kPa}$

$V = ?$

$n = ?$

$R = 8.314\,\text{J K}^{-1}\,\text{mol}^{-1}$

$T = 293\,\text{K}$

$n = \frac{m}{M}$ $n(CaCO_3) = 10.01/100.1 = 0.10\,\text{mol}$

$V = \frac{0.10 \times 8.314 \times 293}{110} = 2.2\,\text{L}$

Chapter review questions

1 **a** Ideal gas is a gas that obeys the relationship linking volume, mass pressure and temperature perfectly. The ideal gas obeys the ideal gas law $PV = nRT$. The real gas is the actual gas being examined.

b At high pressures and low temperatures. And for gases that are highly polar like ammonia.

2 C

3 A

4 C

5 A

6 A

7 B

8 **a** True

b True

c False. At constant temperature and pressure, the volume of a gas increases as the number of moles of the gas increase.

d False. $T_2 = T_1 \times \frac{P_2 \times V_2}{P_1 \times V_1}$

9 $= \frac{V_1}{T_1} = \frac{V_2}{T_2}$

$V_1 = 4.6\,\text{L}$ $\quad V_2 = ?$

$T_1 = 27°\text{C}$ $\quad T_2 = 120°\text{C}$

$= 300\,\text{K}$ $\quad = 393\,\text{K}$

$$V_2 = \frac{4.6 \times 393}{300} = 6.0\,\text{L}$$

10 B

11 A

12 83.3 kPa

13 $PV = nRT$

$P = 1500\,\text{kPa}$

$V = 2\,\text{L}$

$n = ?$

$R = 8.314\,\text{J}\,\text{K}^{-1}\,\text{mol}^{-1}$

$T = 273 + 42 = 315\,\text{K}$

$$n = \frac{1500 \times 2}{8.314 \times 315} = 1.15\,\text{mol}$$

$$n = \frac{n}{M},\ m = nM = 1.15 \times 28 = 32\,\text{g}$$

14 $PV = nRT$

$P = 105.6\,\text{kPa}$

$V = 350\,\text{mL} = 0.350\,\text{L}$

$n = ?$

$R = 8.314\,\text{J}\,\text{K}^{-1}\,\text{mol}^{-1}$

$T = 273 + 26°\text{C} = 299\,\text{K}$

$$n = \frac{105.6 \times 0.350}{8.314 \times 299} = 0.015\,\text{mol}$$

$$n = \frac{m}{M}$$

$$M = \frac{m}{n} = \frac{0.35}{0.015} = 23.5\,\text{g}\,\text{mol}^{-1}$$

15 $PV = nRT$

$P = 1.00\,\text{atm}$

$V = ?$

$n = ?$

$R = 0.082\,\text{L atm}$

$T = 298\,\text{K}$

$$n = \frac{m}{M} \quad n(\text{CO}) = \frac{28}{28} = 1.0\,\text{mol} \quad n(\text{O}_2) = \frac{1}{2}\,n(\text{CO}) = 0.5\,\text{mol}$$

$$V = \frac{0.5 \times 0.082 \times 298}{1} = 12.2\,\text{L}$$

16 $n = \dfrac{m}{M}$

$$n(\text{NH}_4\text{Cl}) = \frac{107}{53.5} = 2.0\,\text{mol}$$

$n(\text{NH}_3) = n(\text{NH}_4\text{Cl})$

$$n = \frac{V}{V_\text{m}},\ V = nV_\text{m} = 2.0 \times 22.4 = 44.8\,\text{L}$$

17 $PV = nRT$

$P = 0.100\,\text{atm}$

$V = 244.5\,\text{L}$

$n = ?$

$R = 0.082\,\text{L atm}$

$T = 273 + 25°\text{C} = 298\,\text{K}$

$$n = \frac{0.100 \times 244.5}{0.082 \times 298} = 1.00\,\text{mol}$$

Using second data point, n calculated to be:

$$n = \frac{0.200 \times 122.2}{0.082 \times 298} = 1.00\,\text{mol}$$

P(atm)	**0.4000**	**2.000**	**8.000**
Calculated V (L)	$V = \frac{1.0 \times 0.082 \times 298}{0.4000}$ $= 61.09\,\text{L}$	$V = \frac{1.0 \times 0.082 \times 298}{2.000}$ $= 12.2\,\text{L}$	$V = \frac{1.0 \times 0.082 \times 298}{8.000}$ $= 3.05\,\text{L}$

18 m(nitroglycerine): density $= 1.59\,\text{g mL}^{-1}$ and volume $= 1\,\text{mL}$ then $m = 1.59\,\text{g}$

$n = \dfrac{m}{M} = 1.59/227 = 0.00070\,\text{mol}$

There are 29 mol gases produced from each 4 mol of nitroglycerine but at standard temperature (273 K) the water is not vapour, so 19 mol gases from each 4 = 4.75 times. Mol of gases produced is $4.75 \times 0.00070 = 0.0033\,\text{mol}$.

Assuming $V = 1\,\text{mL} = 0.001\,\text{L}$

$PV = nRT$

$P = ?$

$V = 0.001\,\text{L}$

$n = 0.0033\,\text{mol}$

$R = 0.082\,\text{L atm}$

$T = 0°\text{C} = 273\,\text{K}$

$P = \dfrac{0.0033 \times 0.082 \times 273}{0.001} = 74.4\,\text{atm}$

19 a kg produced = rate × distance $= 0.16 \times 20\,000 = 3200\,\text{kg}$ in one year

b Amount for one vehicle $\times 10^9 = 3.2 \times 10^{12}\,\text{kg}$ for all vehicles $= 3.2 \times 10^{15}\,\text{g}$

c $PV = nRT$

$P = 1.0\,\text{atm}$

$V = 1.8 \times 10^{18}\,\text{L}$

$n = ?$

$R = 0.082\,\text{L atm}$

$T = 273\,\text{K}$

$n = \dfrac{1.0 \times 1.8 \times 10^{18}}{0.082 \times 273} = 8.0 \times 10^{16}\,\text{mol}$

$m = nM = 8.0 \times 10^{16} \times 44 = 3.5 \times 10^{18}\,\text{g}$

d Fraction $= 3.2 \times 10^{15}/3.5 \times 10^{18} = 9 \times 10^{-4}$

e $3.2 \times 10^{12}\,\text{kg}$ total amount emitted. One fully grown tree takes up 300 kg, then $\dfrac{3.2 \times 10^{12}}{300} =$ 1×10^{10} trees